



WORKSHOP GUIDE

VirtualGlove Controller Enclosure Guide

Print the UNO Q Case, Dock V1, or enclosed Dock V2.1.

2026 EDITION / IAIN BENNETT / MIT LICENSE

VirtualGlove includes three supported enclosure choices: the compact UNO Q Case, the original low-profile Controller Dock V1, and the photo-validated, fully enclosed Controller Dock V2.1. Each design solves a different installation problem. All were designed for an Anycubic Kobra 3, ordinary PLA, a 0.4 mm nozzle, and the Arduino USB-C Hub (8 in 1).

The files are intentionally parametric. Print the small fit tests first, record what changed, and adjust the clearances before committing to a long print.

Choose your case

Choose the enclosure around where the hub should live and which ports must be reachable. After choosing the structure, choose either the small hand/target emblem lid or the larger VirtualGlove wordmark lid. Both lid styles are available for every enclosure.

UNO Q Case



UNO Q Case exterior

Choose the compact case when the Arduino hub will sit elsewhere in the arcade panel. The case protects the UNO Q, exposes only its USB-C connection, keeps the Matrix visible, and leaves airflow around the board.

- Outside size: approximately **90 x 76 x 27 mm** when assembled.
- Best for: a hidden Controller, short cable runs, or the smallest possible box.
- The USB-C hub, camera cable, power cable, and optional Ethernet cable remain outside the enclosure.

Controller Dock V2.1 - revised enclosed hub

Choose V2.1 when the hub should be protected inside a fully closed case. It provides rear USB-C access, a side Ethernet opening, internal USB-A routing, and the clearances measured from the first physical enclosed-dock assembly.



Controller Dock V2.1 exterior

- Outside size: approximately **172 x 126 x 40.4 mm** when assembled.
- The UNO Q moves right to create a broad cable-loop bay on its left.
- Low guides preserve a gentle bend in the hub's thick captive USB-C lead without clamping it.
- Slimmer board posts avoid the UNO Q's underside hardware.
- The base provides approximately **10.5 mm** of free height above the hub before the lid, leaving more room for real plug bodies and cable overmoulds.
- A shorter lid skirt and enlarged corner reliefs prevent the lid lip from binding on the insert towers.
- The full-wordmark recess grows to **102 x 32 mm**, giving the maintained 100 x 30 mm logo a practical printed clearance.
- Separate board-mount and wordmark-recess coupons let you confirm the two corrected fits before the long base or lid print.

Controller Dock V1 - original low-profile dock



Controller Dock exterior

Choose V1 when the lower profile and open service bay suit a fixed installation. Its hub sits low in the rear bay and remains visible and easy to remove. The open top works well with a known cable arrangement, but it does not guarantee normal plug-body clearance on both opposing long port faces at once.

- Outside size: approximately **148 x 104 x 31 mm** when assembled.
- Best for: fixed cable arrangements, pre-connected cables, or adapting an existing V1 print.
- The hub remains removable; it is not trapped inside a sealed hot box.

Compare	UNO Q Case	Dock V1	Dock V2.1
Hub location	Outside	Open rear bay	Enclosed
Port access	UNO Q USB-C	Open top	Rear USB-C, side Ethernet, internal routing
Overall height	About 27 mm	About 31 mm	About 40.4 mm
Lid choices	Small or wordmark	Small or wordmark	Small or wordmark
Best fit	Smallest box	Fixed layout	Protected hub and flexible camera cabling

For assembly at the workbench, use the [eight-page Enclosure Assembly Quick Reference](#). It provides one uninterrupted, model-accurate procedure for each enclosure. This guide adds print settings, fit checks, the reason behind important steps, and troubleshooting.

What you need

VirtualGlove supplies the digital print files. The UNO Q, hub, fasteners, inserts, tools, cables, adhesive, and other physical parts are not included.

For the least ambiguous download, choose the complete bundle for the enclosure you are printing. Existing individual STL and 3MF links remain available below:

- [UNO Q Case print-file bundle](#)
- [Controller Dock V1 print-file bundle](#)
- [Controller Dock V2.1 print-file bundle](#)

Each bundle separates structural parts, matching branding, fit tests, finishing, and parametric source. Read its [PARTS.txt](#) before slicing. The optional large target badge is not included because it does not fit a lid recess.

Fit coupons are test prints, not enclosure pieces. They reproduce only a small opening, cradle, mount pattern, or logo recess so you can try your real hardware before printing a full part. A coupon does not snap into the base or lid; set it aside after the fit check. For the compact UNO Q Case, only the base and one matching lid are structural prints. Its USB-C coupon tests the hub plug, while the hub itself stays outside the case.

Shared hardware and tools

Item	Quantity	Purpose
Arduino UNO Q	1	Runs VirtualGlove
Arduino USB-C Hub (8 in 1)	1	Camera, power, and optional Ethernet; external with the compact case
M3 x 8 mm board screws	4	Fasten the UNO Q to its standoffs
M3 x 8-12 mm case screws	4	Fasten the lid to the base
M3 heat-set inserts	4	Approximately 4.0-4.2 mm outside diameter; receive the case screws
Heat-set insert tool	1	Seats the inserts squarely
M3 driver	1	Installs the board and case screws
Adhesive rubber feet	4	Keeps the enclosure stable and allows bottom airflow

UNO Q Case print set

- [UNO Q base](#)
- Choose one lid: [small-emblem inset](#) or [full-wordmark inset](#)
- Optional print-first test: [USB-C opening coupon](#) (not installed in the case)

The hub is still required, but it remains outside this enclosure and connects through the broad USB-C opening.

Controller Dock V2.1 print set - revised enclosed hub

- [Dock V2.1 base](#)
- Choose one lid: [small-emblem inset](#) or [full-wordmark inset](#)
- [USB-C opening coupon](#)
- [Hub fit coupon](#)
- [UNO Q mount-pattern coupon](#)
- [Full-wordmark recess coupon](#)

Print the two V2.1-specific coupons first. The mount coupon must sit against the UNO Q without a post touching an underside component. The maintained 100 x 30 mm wordmark backing must drop into the recess coupon without bending or force.

Controller Dock V1 print set - original low-profile dock

- [Dock V1 low-profile base](#)
- Choose one lid: [small-emblem inset](#) or [full-wordmark inset](#)
- [USB-C fit coupon](#)
- [Hub fit coupon](#)

Shared finishing parts

The enclosure works without decorative pieces. The Matrix bezel is recommended for a finished edge around the display; choose one emblem only if wanted.

Finishing part	Downloads
Matrix bezel	Cyan bezel
Small lid emblem	Separate: dark backing , cyan hand and target , red target beam ; ACE: single multicolour 3MF
Compact-case wordmark	For the UNO Q full-wordmark lid only. Separate: inset backing , cyan hand and wordmark , red accents ; ACE: single multicolour 3MF
Full-size dock wordmark	Fits the full-wordmark lid for Dock V1 and V2.1. Separate: inset backing , cyan hand and wordmark , red accents ; ACE: single multicolour 3MF
Optional larger emblem	Separate: inset backing , cyan hand and target , red target beam ; ACE: single multicolour 3MF

Every backing now has shallow locating pockets for its cyan and red inserts. The inserts sit nearly flush instead of relying on eye alignment on a flat plate. The small-emblem lids and full-wordmark lids have matching outer recesses, so the completed logo locates positively in the lid. On the compact case and Dock V1, the full-wordmark lid moves its ventilation slots clear of the larger recess; it does not simply cover the original vents.

Lid-branding assembly order

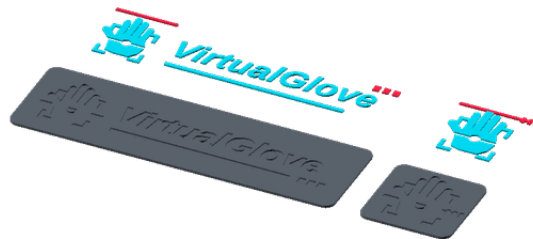
Choose exactly one lid style and its matching logo set:

1. Match the **small-emblem backing** to a small-emblem lid. For a wordmark lid, use the **compact wordmark** on the UNO Q Case or the **full-size wordmark** on Dock V1 and Dock V2.1.
2. Dry-fit the dark backing in the lid recess. It should sit flat without covering the Matrix opening or ventilation slots.
3. Dry-fit the cyan artwork and red accent in the backing's locating pockets. Each piece should sit nearly flush and must not require force.
4. Remove the pieces, apply only a small amount of adhesive, and assemble the cyan and red pieces into the backing.
5. Fit the completed logo set into the lid recess, then fit the separate Matrix bezel around the display opening.

The larger hand/target emblem is a separate decorative part and does not fit a lid recess. The 100 x 30 mm full-size wordmark fits the matching Dock V1 and Dock V2.1 lid recesses directly. It remains too wide for the compact UNO Q Case, which uses the 76 x 22.8 mm compact wordmark.

The 3MF downloads contain the backing, cyan artwork, and red artwork in one file with material colours assigned. In Anycubic Slicer, verify that those three colours map to the intended ACE spools before slicing. The separate STL sets remain available for single-colour printing and hand assembly.

The full-size plaque 3MF is also saved as an Anycubic Slicer Next project for the four-spool order used during development: **1 black, 2 purple, 3 red, 4 cyan**. Black prints the backing, red prints the small accents, and cyan prints the hand, target, and wordmark. Purple is intentionally loaded but unused.



Recessed logo backings with cyan and red inserts separated for assembly



Small-emblem and full-wordmark lid inset options

Optional materials and connections

- Thin double-sided adhesive or plastic-safe glue for the branding layers.
- Approximately 1 mm foam or TPU strips beneath the Dock's hub if it needs a quieter or firmer fit.
- A camera connected to USB-A 3.0 or the USB-C data port on the hub.
- USB-C PD power connected to the hub.
- Ethernet connected through Dock V2.1's right-side opening, or directly to the external hub used with the compact case.

The hub's connector shape does not guarantee its speed. USB-A 3.0 provides up to 5 Gbps, while this hub's USB-C data port provides 480 Mbps. A camera that requires USB 3 bandwidth should use USB-A 3.0, with an appropriate cable or adapter when the camera itself has a USB-C socket. Never connect the camera to the USB-C PD power input.

The full-logo and hand/target pieces are printable adaptations of the project artwork. They preserve the italic wordmark, hand, target, cyan linework, and red accents while removing screen-only glow and details smaller than a dependable 0.4 mm nozzle can reproduce.



Print-safe full logo and hand-target emblem

To change a dimension, download the [parametric OpenSCAD source](#) and the [STL export script](#).

Prepare and print

Current fit revision

Use a base and lid from the same named design. Dock V2.1 corrects four first-print problems found with the real UNO Q, Arduino hub, and captive cable:

- the UNO Q supports use 4.8 mm tops and 5.8 mm feet to clear underside parts;
- the UNO Q moves farther right, leaving a broad loop instead of folding the hub cable tightly beside the left wall;
- low cable guides retain that loop without clamping it; and
- the lid uses a 3.2 mm skirt, 11 mm corner reliefs, and extra vertical space so it closes on the rim rather than the screw posts, hub, or plug bodies.

If you printed a base or lid from an earlier download, replace both structural pieces with the current matching files before judging the fit. Do not trim a corner boss or force the lid over it.

First check-in: print the coupons

Print the [USB-C fit coupon](#) for any enclosure. If building a Dock, also print the [hub fit coupon](#).

For V2.1, also print its [mount-pattern coupon](#) and [wordmark-recess coupon](#).

The USB-C plug should enter without scraping and should not have enough side play to pull hard against the UNO Q socket. The hub should sit in its channel without bowing the Dock coupon or rattling loosely.

For Dock V2.1, also place the real hub on the unpowered printed base before installing electronics. Confirm that it rests in the low cradle, its USB-C connectors align with the broad rear access opening, its Ethernet socket aligns with the right-side opening, and the low rails remain below every connector. Fit the intended USB-A camera cable internally and make sure its cable-not its plug-can turn around the hub end and leave through a rear routing opening while the lid closes without pressure.

Record the printer, material, nozzle, slicer profile, and any dimensional change you make. In [virtualglove-controller.scad](#), [fit](#) controls ordinary mating clearance and [hub_fit](#) controls the hub cradle. Do not scale the entire model to fix one opening; change the relevant clearance instead.

Kobra 3 PLA starting profile

- 0.4 mm nozzle
- 0.20 mm layer height
- four walls
- five top and bottom layers
- 20-25% gyroid infill
- supports off initially
- brim only if the base corners lift
- base and lid in the supplied orientation

The lid's visible face lies on the bed for a clean finish. Dock V2.1's internal hub cradle prints as part of the base and does not require support. PLA is suitable inside a normal arcade cabinet. Do not leave the Controller in a hot car or another high-temperature location, and never cover its ventilation slots.

Dimensions and reference drawings

The design follows Arduino's published **68.58 x 53.34 mm** UNO Q outline and four mounting holes. The hub envelope is **119 x 27.8 x 16 mm** with a fixed 175 mm cable. Arduino does not publish every plug-body and overmould dimension, which is why the fit coupons are part of the build rather than an optional extra.

The UNO Q's bottom-side parts remain below 2 mm according to Arduino. The mounted board has 4.6 mm of clear height above the enclosure floor. Each lid has the same outside width and depth as its base. V1 uses the original 6 mm skirt and 9.2 mm boss reliefs. V2.1 uses a 3.2 mm skirt and 11 mm reliefs, leaving 1.5 mm radial clearance around each 8 mm corner boss. These features let the top panel land flat on the full case rim.

- [Arduino UNO Q datasheet](#)
- [Arduino UNO Q STEP model](#)
- [Arduino USB-C Hub datasheet](#)

Printing the branding

The lid emblem is a compact version of the hand/target project mark. For the closest match to the VirtualGlove artwork, use charcoal or black for the backing, cyan for the hand and target corners, and red for the target beam. The larger optional plaque combines the same mark with the full wordmark.

For an ACE multicolour print, open the matching 3MF and confirm the charcoal, cyan, and red material assignments. For separate printing, use the three STL files, test each insert in its backing pocket, and glue only after the fit is correct. The pockets provide alignment; do not force an insert that has swollen from first-layer expansion. Separate accents avoid wasting material on colour changes through an entire enclosure.

Use a small amount of thin double-sided adhesive or plastic-safe glue. Dry-fit the pieces first; the detailed layers are intentionally shallow and should not be forced.

Understand and assemble

Rear and side views

Left and right are named while looking at the Controller from the front/Matrix side. The coloured connector blocks in the V2.1 illustrations identify the hidden hub; they are not additional printed parts. Red marks USB-C PD power, cyan marks USB-C data or USB-A 3.0, and silver represents other hub connectors, including Ethernet.

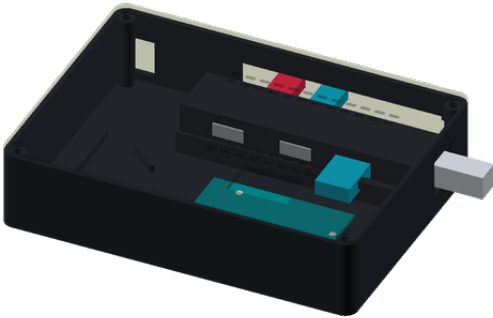
Controller Dock V2.1

The hub sits inside the rear of the closed case. The rear view shows direct access to its outward-facing USB-C bank plus cable-routing openings at the corners. The cutaway shows how a cable plugged into the inward-facing USB-A bank turns around the hub and exits at the rear. The right view shows the separate Ethernet opening.

View	Controller Dock V2.1	What to check
Front		Lid closes over both the UNO Q and hub.
Rear		USB-C PD and data remain directly pluggable; routed cables leave through the corner openings.
Left		No hub body or loose connector projects above the lid.

View	Controller Dock V2.1	What to check
Right		Ethernet plugs directly into the hidden hub through the side opening.

The V2.1 port-access view deliberately makes the lid translucent. It shows representative USB-C plugs entering from the rear and a USB-A plug fitted inside. It verifies the intended routing concept, not the exact shape or bend radius of every third-party cable.



Controller Dock V2.1 cutaway with rear and internally routed connections

UNO Q Case and Controller Dock V1

The UNO Q Case exposes the broad USB-C opening. Dock V1 uses the original open, low rear service bay. Both are complete, supported alternatives rather than historical diagrams included only for reference.

View	UNO Q Case	Controller Dock V1
Rear		

View	UNO Q Case	Controller Dock V1
Left		
Right		

Exploded assembly

These views show the assembly order rather than print orientation. Fasteners and heat-set inserts are omitted so the main layers remain easy to see. In V2.1, the UNO Q and hub both seat in the base before the lid closes over them. The original V1 hub instead remains visible in its low open rear bay.



Exploded assembly view of Controller Dock V2.1



Keep the [eight-page Enclosure Assembly Quick Reference](#) beside the workbench for the model-accurate sequence. Page 3 covers the UNO Q Case, pages 4-5 cover Dock V1, pages 6-7 cover Dock V2.1, and page 8 is the shared inspection. The detailed steps below use the same numbers and explain the fit checks that accompany each action.

Assemble the UNO Q Case

These six steps match page 3 of the Assembly Quick Reference.

- 1. Seat the four inserts.** Heat the four inserts into the corner bosses. Keep them square and stop when they are flush; excess heat can soften the boss.
- 2. Place the UNO Q.** Set it on the four standoffs with the USB-C socket facing the broad opening.
- 3. Fasten the board.** Use four M3 x 8 mm screws. Tighten only until the board is secure and remains flat.
- 4. Check USB-C clearance.** Connect the Arduino hub through the opening and confirm its plug does not press sideways on the socket.
- 5. Choose and dry-fit the branding.** Match one lid logo set to the printed lid. Dry-fit its backing, cyan artwork, and red accent, and confirm that the Matrix opening, ventilation slots, and separate bezel remain clear.
- 6. Close and add feet.** Fasten the lid and attach four rubber feet, then complete the shared pre-power inspection.

Only the UNO Q USB-C port is intentionally exposed. Do not force a thick plug through the opening; adjust and reprint the coupon if necessary.

Assemble Controller Dock V2.1

These eight steps match pages 6-7 of the Assembly Quick Reference. Check the two V2.1-specific coupons before step 1.

- 1. Seat the four inserts.** Heat them into the corner bosses, keeping them square and flush.
- 2. Mount the UNO Q.** Place it with the USB-C socket facing the broad opening, then fasten it without bending the board.
- 3. Orient the hidden hub.** Seat it in the rear cradle with its USB-C PD and data bank facing the broad rear opening. Route its captive cable internally to the UNO Q without twisting the socket. The Ethernet end sits beside the right wall; the open bay at the hub's left end is reserved for the thick captive cable.
- 4. Connect the captive cable.** Route the hub lead through its internal channel to the UNO Q without pulling or twisting either connector.

5. **Choose the camera route.** If the camera uses USB-A, connect it to the inward-facing USB-A 3.0 port now. Turn the flexible cable-not the plug-around the left hub end and out a rear routing opening.
6. **Keep power and Ethernet clear.** If the camera uses USB-C, leave its data port accessible through the rear bank. Keep the neighbouring USB-C PD power input accessible as well. If using Ethernet, confirm the RJ45 socket faces the right-side opening.
7. **Dry-fit the lid.** Confirm it closes completely without touching the hub, plugs, cables, or board and that no cable is pinched at a routing opening. Look through the open base and confirm all four skirt reliefs surround their corner bosses rather than resting on them. Remove the lid, connect any remaining internal cable, and repeat the closure check. Plug rear-accessible USB-C cables in only after the lid is fastened.
8. **Close and finish.** Add the Matrix bezel and the matching assembled lid logo set, fasten the lid, and attach four rubber feet.

The rails locate the hub and are intentionally lower than its connector openings. If a rail or rear opening obstructs a plug, or the lid presses on an internally connected cable, stop and adjust the model rather than forcing it.

Assemble Controller Dock V1

These eight steps match pages 4-5 of the Assembly Quick Reference.

1. **Seat the four inserts.** Heat them into the corner bosses, keeping them square and flush.
2. **Place the UNO Q.** Set it on its standoffs with the USB-C socket facing the broad opening.
3. **Fasten the board.** Use four M3 x 8 mm screws and keep the board flat.
4. **Add optional cradle padding.** Fit thin foam or TPU strips to the low hub cradle only if desired.
5. **Connect the captive cable.** Route it through the internal channel and connect it to the UNO Q.
6. **Seat the hub.** Lower it into the open rear bay without trapping the captive lead.
7. **Connect the fixed cable set.** Confirm the ports required by this arrangement, the Ethernet end, and the captive lead are free. Connect the camera to a USB-A 3.0 port, power to the hub's USB-C PD input, and optional Ethernet.
8. **Close and finish.** Fit the lid, Matrix bezel, and the matching assembled lid logo set, then add four rubber feet.

Second check-in: inspect before power

Before switching on the Controller, confirm:

- the board is flat and no standoff is twisting it;
- screw heads and inserts cannot touch components or solder joints;
- the hub and USB-C cable are not pushing sideways on the UNO Q socket;
- the Matrix window and every ventilation opening are clear;
- no loose screw, wire strand, or printed debris remains inside; and
- the lid closes without pressure on the board, hub, or cables.

Power the assembled Controller for 30 minutes with the camera active. Check that the case remains comfortable to touch, the camera stays connected, and the Matrix and Dashboard behave normally. A warm board is expected; a softening PLA case, repeated USB disconnect, or unusually hot enclosure is not.

Customise and maintain

Make a revision

Open [virtualglove-controller.scad](#) in OpenSCAD. The dimensions and clearances that are most likely to need adjustment are grouped at the top of the file. Choose a part from the [part](#) selector, preview it, and export it directly, or regenerate the complete checked-in STL set with:

```
BASH
hardware/enclosures/export-stl.sh
```

Keep the source file, STL files, and this guide together in a check-in. Note the printer and measured fit when changing a hardware dimension so the next person can tell whether the change is universal or printer-specific.

Safety and limits

- Disconnect power before opening the case.
- Handle the UNO Q with normal electrostatic-discharge care.
- Never force the board, hub, plug, insert, or lid into place.
- Keep metal fasteners and decorative foil away from antennas and electronics.
- Keep airflow paths open and inspect PLA periodically in a warm cabinet.
- Treat this as a first-fit printable design until it has been measured against the exact hub, plugs, inserts, and printer that will be used.

The enclosure protects and organises the Controller; it does not make the UNO Q or hub waterproof, impact-rated, or suitable for outdoor use.